

W

hat's the first thing that comes to mind when you think of an astronaut? The Spacesuit of course!

The dorky yet totally awesome fluffy white thing with a fishbowl head has become synonymous with space and space travel. They evoke feelings of awe and envy among kids (and quite a few adults) and bring about a longing to explore one of the final frontiers left for humankind as a whole. But have you really wondered about what the spacesuit actually is and what it does? And there has been a gradual but steady evolution in design, bringing both form and function.

WHAT IS A SPACESUIT?

A spacesuit, when you boil it down, is simply a mobile, self-contained space-ship for one person. It has to keep the person inside alive, whether they are outside in the vacuum of space or inside the spacecraft. It's used for deploying payloads, inspecting and repairing spacecraft and satellites, retrieving satellites and even construction of space stations. It has to regulate temperature while withstanding external temperatures of -450 degrees to 250 degrees Fahrenheit, provide air as well filter out CO₂, have a communications array along with various sensors, extract cough excreta, maintain stable pressure, protect the astronaut against micrometeoroids and loads of other functions. All of this is handled by various devices that are all rather seamlessly integrated into the suit itself.

There are three major classifications of a spacesuit including IntraVehicular Activity or IVA, ExtraVehicular Activity or EVA and Intra/ExtraVehicular Activity or IEVA.

IVA suits are worn inside the spacecraft just for the risk of loss of cabin pressure. They are generally very comfortable to wear and relatively lightweight. EVA suits are meant for spacewalks and other outer excursions while offering maximum protection. As a result, they are the heaviest and bulkiest. IEVA suits are for situations where you will not be outside in space



Spacesuits of the future

One of the most iconic symbols of Space Travel and it's decidedly awesome evolution.

Prithvi Sudhan | feedback@digit.in

but aren't in the pressurised aircraft (like when you're repairing an airlock) and so, are a mixture of IVA and EVA.

HISTORY OF THE SPACE SUIT

The very first spacesuits were exclusively IVAs, as suits that could reliably stand up to the conditions outside in the vacuum of space were a bit difficult for engineers to design. This was because they had no reference to what effect things like zero gravity had on the human body. In fact, the first IVA designs for suborbital flight were merely adapted deep sea suits and high altitude suits. There were plenty of theories for spacesuit design from soft suits made from all fabric (scrapped due to not be-

ing able to hold pressure well) to all hard material suits (Mecha armor, anyone?) which was scrapped due to extremely limited mobility and dexterity. A good balance of the two had to be found, incorporating the physical support and protection of a hard suit while retaining most of the mobility of a soft suit.

In 1961, Yuri Gagarin, the first man in space also wore the first space suit; "Skafandr Kosmicheskii" #1 or SK-1. It was custom made for him and weighed a whopping 44 pounds, mainly accounting for the protection gear in the case of an ejection. The Russians didn't really have a landing program per se so they designed the suit to protect Gagarin for rough falls (upto 8 km!) A month later,